

30-Minute Student Task

Build Your Own Rule-Based Python Chatbot

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Programming Language	Python
Assessment	Rule-Based Conversational Chatbot

Assessment

A simple rule-based conversational chatbot was developed in Python. The chatbot accepts user input continuously, recognizes greetings and different questions, responds accordingly, and exits when the user types **bye** or **exit**.

Chatbot Features

- Accepts user input continuously using the input() function.
- Recognizes common greetings such as hi, hello, and hey.
- Responds to simple questions such as how are you and your name.
- Provides a help response when the user asks for help.
- Handles unknown questions with a default response.
- Terminates the conversation when the user enters bye or exit.

Python Chatbot Implementation

Python program implementing the rule-based conversational chatbot.

```
def chatbot():
    print("Chatbot: Hello! I am a simple chatbot. How can I help you?")

    while True:
        user_input = input("You: ").lower().strip()

        if user_input in ["hi", "hello", "hey"]:
            print("Chatbot: Hello! Nice to meet you.")

        elif "how are you" in user_input:
            print("Chatbot: I'm doing great! How are you?")

        elif "your name" in user_input:
            print("Chatbot: My name is SimpleBot.")

        elif "help" in user_input:
            print("Chatbot: Sure! I can answer simple questions and greetings.")

        elif user_input in ["bye", "exit"]:
            print("Chatbot: Goodbye! Have a nice day.")
            break

        elif "thank" in user_input:
            print("Chatbot: You're welcome!")

        else:
            print("Chatbot: Sorry, I don't understand that.")

chatbot()
```

The program uses simple if-elif-else rules to match the user's message with a predefined response.

Chatbot Execution and Test Output

Sample terminal output showing greeting, question handling, an unknown message, and the exit command.

```
Chatbot: Hello! I am a simple chatbot. How can I help you?  
You: hello  
Chatbot: Hello! Nice to meet you.  
You: what is your name  
Chatbot: My name is SimpleBot.  
You: how are you  
Chatbot: I'm doing great! How are you?  
You: bye  
Chatbot: Goodbye! Have a nice day.
```

Result

The chatbot successfully accepts continuous user input, identifies predefined conversational patterns, provides appropriate responses, and stops the program when the user enters bye or exit.